[SCHOOL NAME — PLACEHOLDER LETTERHEAD]

(Swap in your school's actual letterhead / logo here)

Class 4 Maths — CA-1 (Hard) — Answer Key

Class : 4 | Max Marks : 70

Subject : Maths (Math-Mela) | Duration : 2 Hr 30 Min

Syllabus: Chapters 1–3 — Shapes Around Us, Hide and Seek, Pattern Around Us

General Instructions: All questions are compulsory. For Section A, write the correct option letter.

ANSWER KEY

SECTION A — Multiple Choice Questions (1 Mark each)

1. A solid shape has 5 faces, 9 edges, and 6 corners. Using Faces + Corners - Edges = 2, verify if this could be a valid solid, and identify it. *(1 Mark)*

A. 5 + 6 - 9 = 2, valid; likely a triangular prism ✓
B. 5 + 6 - 9 = 0, invalid
C. 5 + 9 - 6 = 8, invalid
D. The formula cannot be used here

2. If you see only the top view and front view of an object but not the side view, what limitation might you face in fully understanding its shape? *(1 Mark)*

A. No limitation, two views are always enough
B. You might not know the exact depth or shape variations only visible from the side ✓
C. Top and front views are identical, so no information is lost
D. It becomes impossible to know anything about the object

3. A sequence of numbers goes 2, 4, 6, 8, ... What is the 10th number in this pattern? *(1 Mark)*

A. 18
B. 20 ✓
C. 22
D. 16

4. Which of these combinations of coins/notes to make ₹145 uses the FEWEST number of pieces (available: 100, 50, 20, 10, 5)? *(1 Mark)*

A. 100 + 20 + 20 + 5 ✓
B. 50 + 50 + 20 + 20 + 5
C. 100 + 50 - 5 (not valid)
D. 20 x 7 + 5

5. A pyramid has a square base. How many triangular faces will it have in addition to its base? *(1 Mark)*

A. 2
B. 3
C. 4 ✓
D. 5

6. If a grid position is described as "2 rows down and 3 columns right" from the top-left corner, which of these best describes this position? *(1 Mark)*

A. Row 2, Column 3 ✓
B. Row 3, Column 2
C. Row 1, Column 1
D. It cannot be determined

7. Which statement is TRUE about the sum of two odd numbers? *(1 Mark)*

A. The sum is always odd
B. The sum is always even ✓
C. The sum can be either odd or even
D. Odd numbers cannot be added together

8. A cube-shaped box is cut in half straight through the middle, parallel to one face. What shape will the cut cross-section be? *(1 Mark)*

A. Circle
B. Square ✓
C. Triangle
D. Pentagon

9. If Priya has ₹350 and wants to buy an item worth ₹275, how much change should she receive, and using the fewest notes/coins (100, 50, 20, 10, 5), how could this change be given? *(1 Mark)*

A. ₹75 as 50+20+5 ✓
B. ₹85 as 50+20+10+5
C. ₹75 as 20x3+10+5
D. ₹65 as 50+10+5

SECTION B — Short Answer Questions (2 Marks each)

1. Explain why knowing only the top view of a building is not enough to fully construct it, using reasoning about missing information. *(0 Marks)*

➤ The top view only shows the layout and outline of the building from above, but it does not show the height of walls, the design of the front entrance, or other vertical details, all of which are necessary for actually constructing the building, so front and side views are also needed.

2. A pattern of numbers goes 3, 7, 11, 15, ... Identify the rule of the pattern and find the next two numbers. *(0 Marks)*

➤ The rule is adding 4 to the previous number each time (3+4=7, 7+4=11, 11+4=15); following this rule, the next two numbers are 19 and 23.

3. Explain why a triangular pyramid (with a triangular base) has fewer faces than a square pyramid (with a square base), using their properties. *(0 Marks)*

➤ A triangular pyramid has 1 triangular base + 3 triangular side faces = 4 faces total, while a square pyramid has 1 square base + 4 triangular side faces = 5 faces total; the difference comes from the base shape needing more side faces to enclose a square base than a triangular one.

4. A group of 47 students is being divided for two different games - one needing pairs, and one needing groups of 5. Explain which grouping will have students left over and why. *(0 Marks)*

➤ For pairs (groups of 2), 47 is odd, so 23 pairs can be formed with 1 student left over; for groups of 5, 47 ÷ 5 = 9 remainder 2, so 9 groups of 5 can be formed with 2 students left over - both groupings leave some students without a complete group.

5. Explain how understanding both grid positions and directions (like up, down, left, right) together helps in describing movement, such as in a simple maze. *(0 Marks)*

➤ Grid positions tell you exactly where something is located (like row 3, column 2), while directions tell you how to move from one position to another (like "move right two columns"); together, they allow precise navigation instructions, such as solving or describing a path through a maze.

6. A cube net (unfolded cube) is shown with 6 squares arranged in a cross shape. Explain how you would verify that this net can actually fold into a cube. *(0 Marks)*

➤ To verify, you check that there are exactly 6 square faces (matching a cube's 6 faces), and that when mentally folded along the edges, each square would align to form a closed 3D shape with no overlaps or gaps, matching all of a cube's properties.

7. Explain why the statement "all shapes with 4 sides are squares" is incorrect, using at least two counter-examples. *(0 Marks)*

➤ This statement is incorrect because shapes like rectangles (with unequal side lengths) and parallelograms (with slanted sides) also have exactly 4 sides but are not squares, showing that having 4 sides alone does not make a shape a square - a square also needs all sides equal and all angles to be right angles.

8. A vending machine only accepts exact change for a ₹65 item. Explain, using available denominations of ₹50, ₹20, ₹10, ₹5, how many different combinations could make exactly ₹65. *(0 Marks)*

➤ Some combinations include: 50 + 10 + 5 = 65; 50 + 5 + 5 + 5 = 65 (using multiple 5s); 20 + 20 + 20 + 5 = 65; and 20 + 10 + 10 + 10 + 10 + 5 = 65 - showing there are multiple valid ways to reach the exact amount using different combinations of the available denominations.

9. Explain why understanding both 2D shape properties and 3D shape properties together is important when studying real architecture, using an example. *(0 Marks)*

➤ 2D shape knowledge helps understand blueprints and floor plans (flat representations), while 3D shape knowledge helps understand the actual structure being built, like how a 2D rectangular floor plan translates into a 3D cuboid-shaped room - both types of knowledge are needed to fully understand and design real buildings.

10. A number pattern skips every third number: 1, 2, 4, 5, 7, 8, ... Explain the rule and predict the next three numbers. *(0 Marks)*

➤ The rule skips multiples of 3 (3, 6, 9, ...) from the counting sequence; following the pattern (1,2 skip3, 4,5 skip6, 7,8 skip9...), the next three numbers would be 10, 11, and then skip 12, giving 10, 11, 13.

11. Explain the concept of symmetry in relation to shapes like a square, and why a scalene triangle (with all different side lengths) generally lacks this property. *(0 Marks)*

➤ Symmetry means a shape can be divided into matching, mirror-image halves; a square has multiple lines of symmetry because its equal sides and angles create balanced halves, while a scalene triangle, having all different side lengths and angles, usually cannot be split into matching mirror-image halves, so it generally lacks symmetry.

12. Explain how a "bird's eye view" (top view) used in maps differs conceptually from a "cross-sectional view" (a view showing what's inside if something were cut open). *(0 Marks)*

➤ A bird's eye view (top view) shows the external layout of something as seen from above without revealing what is inside, while a cross-sectional view shows the internal structure by imagining the object cut open, so the two views serve different purposes - one for external layout and one for internal structure.

13. Explain why the concept of "fewest coins/notes" when making change is considered an efficient strategy, and describe one situation where this might not be practical. *(0 Marks)*

➤ Using the fewest coins/notes is efficient because it reduces the physical amount of money to handle and count, saving time; however, it might not be practical if the shopkeeper does not have larger denomination notes/coins available, forcing them to use more smaller denominations instead.

14. A 3D shape has exactly one flat face and one curved surface, meeting at a single point. Identify the shape and explain your reasoning. *(0 Marks)*

➤ This describes a cone, since it has one flat circular face (the base) and one curved surface that comes together at a single pointed vertex (the apex) at the top.

SECTION C — Short Answer Questions (Explain) (3 Marks each)

1. Analyse how the properties of 3D shapes (faces, edges, corners) follow a predictable mathematical relationship (Faces + Corners - Edges = 2), and discuss why this rule is useful for verifying whether a described shape is possible. *(0 Marks)*

➤ This rule, known as Euler's formula, shows that the numbers of faces, corners, and edges of many solid shapes are not random but follow a consistent mathematical relationship.

➤ By applying this formula to a described shape, we can check if the given numbers are mathematically possible - for example, a shape with 5 faces, 6 corners, and 9 edges checks out (5+6-9=2), confirming it could be a valid solid like a triangular prism.

➤ This is useful because it allows us to verify shape properties logically, rather than only through physical observation, which is especially helpful when working with descriptions or diagrams rather than real 3D objects.

2. Discuss how combining knowledge of views (top, front, side) with grid-based positioning creates a more complete spatial understanding than either skill alone. *(0 Marks)*

➤ Understanding different views (top, front, side) helps us understand the shape and structure of an object from multiple angles, which is important for recognising 3D objects fully.

➤ Grid-based positioning helps us describe exact locations within a space, which is useful for navigation and organisation, such as finding a specific desk in a classroom.

➤ Combining both skills allows for a genuinely complete spatial understanding - being able to both recognise what something looks like from different angles AND pinpoint exactly where it is located, which is valuable in fields like architecture, mapping, and even video game design.

3. Critically evaluate why understanding even and odd number patterns, though seemingly simple, forms an important foundation for more advanced mathematical concepts learned in later years. *(0 Marks)*

➤ Even and odd numbers introduce the basic idea of number classification based on a rule (divisibility by 2), which is a foundational concept for understanding more complex ideas like factors, multiples, and divisibility rules in later years.

➤ This concept also introduces pattern recognition in a simple, visual way (through pairing), which builds the logical thinking skills needed for algebra and more abstract mathematics later on.

➤ Without a solid understanding of these basic patterns, students might struggle with more advanced topics that build directly on this foundational number sense.

4. Discuss the practical and mathematical value of learning to make payments using the fewest possible notes/coins, connecting this to broader real-world financial literacy. *(0 Marks)*

➤ Mathematically, this skill involves understanding place value and combinations, reinforcing addition and subtraction skills in a practical context.

➤ Practically, this skill teaches efficient money handling, which is a valuable real-world financial literacy skill useful throughout a person's life, whether shopping, budgeting, or managing a small business.

➤ This connection between abstract maths (number combinations) and practical life skills (handling money) shows how mathematics taught in school often has direct, everyday applications beyond the classroom.

5. Analyse how the three chapters (Shapes Around Us, Hide and Seek, Pattern Around Us) together build a foundation for spatial reasoning and logical thinking, which are important for subjects beyond mathematics. *(0 Marks)*

➤ "Shapes Around Us" builds foundational geometric understanding, teaching students to recognise, classify, and analyse the properties of both 2D and 3D shapes.

➤ "Hide and Seek" builds spatial reasoning skills, helping students understand perspective, position, and direction, which are valuable not just in maths but also in subjects like art, geography, and even physical education (understanding space and movement).

➤ "Pattern Around Us" builds logical and numerical thinking through pattern recognition, which supports skills needed in subjects like computer science (recognising sequences/algorithms) and even music (recognising rhythmic patterns).

➤ Together, these three chapters develop a well-rounded foundation of spatial and logical thinking that extends far beyond mathematics alone, supporting learning across many other subjects.

6. Discuss whether learning about 3D shapes primarily through classroom activities (like building models) is more effective than learning purely through textbook diagrams, and explain your reasoning. *(0 Marks)*

➤ Hands-on activities, like building models with blocks or straws, allow students to physically manipulate and observe 3D shapes from all angles, which can build a deeper, more intuitive understanding than flat textbook diagrams alone.

➤ However, textbook diagrams are useful for reinforcing vocabulary, formal definitions, and allowing students to practise identifying shapes in a standardised, exam-relevant format.

➤ A combination of both approaches - hands-on building for intuitive understanding and textbook diagrams for formal reinforcement - is likely most effective, as each method strengthens different aspects of understanding 3D shapes.

SECTION D — Long Answer Questions (5 Marks each)

1. Write a detailed essay analysing how the three chapters studied (Shapes Around Us, Hide and Seek, Pattern Around Us) connect to real-world careers or fields, such as architecture, design, or data analysis, explaining the specific skills from each chapter that would be useful. *(0 Marks)*

➤ "Shapes Around Us" directly connects to careers like architecture and engineering, where understanding the properties of 2D and 3D shapes (faces, edges, corners, angles) is essential for designing stable, functional structures.

➤ "Hide and Seek" connects to fields like architecture, urban planning, and even game design, where understanding different views (top, front, side) and spatial positioning (grids, directions) is crucial for creating accurate plans, maps, or virtual environments.

➤ "Pattern Around Us" connects to fields like data analysis, computer programming, and finance, where recognising numerical patterns, understanding even/odd classifications, and efficiently managing quantities (like money) are foundational skills.

➤ Together, these three chapters, though taught at a basic level in Class 4, lay the groundwork for skills that professionals in many different fields continue to build upon and use throughout their careers, showing the real-world relevance of early mathematics education.

2. Design a small "maths trail" activity for younger students, using at least one concept each from "Shapes Around Us," "Hide and Seek," and "Pattern Around Us." Describe the activity in detail and explain how it teaches each concept. *(0 Marks)*

➤ This is a creative, open-ended design task; there is no single correct answer, but strong responses should clearly incorporate concepts from all three chapters.

➤ A strong answer might design a trail where students first identify and classify shapes found around the school (Shapes Around Us), then follow grid-based directions or compare different views of an object to find their next clue (Hide and Seek), and finally solve an even/odd number riddle or count money to unlock a final prize (Pattern Around Us).

➤ The explanation should clearly connect each part of the activity back to the specific mathematical concept it reinforces, showing a thoughtful integration of geometry, spatial reasoning, and number patterns into one cohesive, engaging activity.

3. Reflect on which of the three chapters (Shapes Around Us, Hide and Seek, Pattern Around Us) you found most interesting or useful, and write a detailed explanation of why, including specific examples from the chapter and how you might use this knowledge in your own life. *(0 Marks)*

➤ This is a personal, open-ended reflective answer, and any of the three chapters can be validly chosen if well-reasoned.

➤ A strong answer should clearly state which chapter was chosen, describe specific concepts learned from it (such as identifying 3D shapes, understanding grid positions, or recognising number patterns), and explain genuine, thoughtful reasons why this chapter was interesting or useful.

➤ The response should also include a realistic example of how this knowledge might be applied in the student's own life, such as using shape knowledge while building something, using view/grid knowledge while reading a map, or using pattern knowledge while managing pocket money.